INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

MEASUREMENT SET, TRANSMISSION IMPAIRMENT

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a hand-held Transmission Impairment Measurement Set (TIMS). This CID is meant as a minimum requirement for the TIMS in which only those manufacturers that meet or surpass the following requirements are to supply the TIMS per this CID. The TIMS makes level, frequency, noise, P/AR (Peak-to-Average-Ratio), and signal-to-noise ratio measurements on 2-wire and 4-wire telephone circuits as well as testing for phase jitter, amplitude jitter, gain hits, phase hits, and dropouts.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet this description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The TIMS shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

2.2.1 Temperature Operating. The TIMS shall conform to its specified performance and accuracy while being operated in a temperature range of 0°C to +40°C.

2.2.3 Temperature Non-operating. The TIMS shall conform to its specified performance and accuracy after being stored in the temperature range of -10°C to +60°C.

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2.3 Electrical Power Sources. The equipment shall operate from an internal, rechargeable battery pack for a minimum of three hours, and shall be equipped with an AC adapter to supply power to an internal recharging circuit. Battery recharge time shall be a maximum of four hours. The AC adapter shall operate from nominal line voltages of 110 volts AC (±10%) and frequencies of 45 Hz to 60 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 1500 hours of operation with a statistical certainty of 95%.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the equipment.

2.6 Performance Requirements. The TIMS shall meet all requirements specified herein, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800. The TIMS shall meet or exceed the following measurement parameters in accordance with IEEE 743-1995, or the equivalent measurement parameters, except as specified herein. IEEE 743-1984 equivalent measurement parameters are acceptable.

2.6.1 Test Set Receiver Specifications.

2.6.1.1 Frequency.

(a) Range	200 Hz to 19,999 Hz
(b) Resolution	1 Hz (200 Hz to 9,999 Hz) 10 Hz (10 kHz to 19,999 Hz)
(c) Accuracy	± 1 Hz (200 Hz to 9,999 Hz) ± 10 Hz (10 kHz to 19,999 Hz)

2.6.1.2 Level.

(a) Range	-49.9 dBm to +10 dBm
(b) Resolution	0.1 dB
(c) Accuracy	± 0.1 dB (-19 dBm to 0 dBm) at a fixed frequency between 1002 Hz to 1020 Hz ± 0.2 dB (-40 dBm to +10 dBm)(200 Hz - 14,999 Hz) ± 0.5 dB (-40 dBm to +10 dBm)(15 kHz - 19,999 Hz) ± 0.4 dB (-49.9 dBm to -40 dBm)(200 Hz - 14,999 Hz) ± 0.8 dB (-49.9 dBm to -40 dBm)(15 kHz - 19,999 Hz)
(d) Detectors	true rms or average

2.6.2 Test Set Transmitter Specifications.

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2.6.2.1 Tone generators, common requirements.

2.6.2.1.1 Level.

(a) Range	-40 dBm to +10 dBm
(b) Resolution	0.1 dB
(c) Accuracy	± 0.2 dB (200 Hz to 14,999 Hz)
	± 0.5 dB (15 kHz to 19,999 Hz)

2.6.2.2 Variable frequency generator.

2.6.2.2.1 Frequency.

(a) Range	200 Hz to 19,999 Hz
(b) Resolution	1 Hz (200 Hz to 9,999 Hz)
	10 Hz (10 kHz to 19,999 Hz)
(c) Accuracy	± 1 Hz (200 Hz to 9,999 Hz)
	± 10 Hz (10 kHz to 19,999 Hz)

2.6.2.2.2 Flatness.

(a) 0.2 dB from 200 Hz to 15 kHz
(b) 0.5 dB elsewhere

2.6.2.2.3 SF Skip. Skip Frequency of 2450 Hz to 2750 Hz.

2.6.2.3 Holding tone generator.

2.6.2.3.1 Frequency.

(a) Range	One or more fixed frequencies from 1002 Hz to 1022 Hz
(b) Accuracy	± 0.005 Hz

2.6.2.3.2 Spurious Noise. Any generator used as a source of holding tone shall have:

(a) 0.2° phase jitter or less
(b) 0.2% amplitude jitter or less
(c) No impulse noise at a level 20 dB below the power of the tone
(d) No phase hits greater than 3°
(e) No gain hits greater than 0.5 dB
(f) A signal-to-total-distortion ratio (S/TD) of at least 50 dB

2.6.3 Test Set Measurement Capabilities.

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2.6.3.1 Noise Measurements.
Transmitter Mode: quiet termination.
Receiver:

(a) Level Range	10 dBrn to 90 dBrn
(b) Resolution	1 dB
(c) Accuracy	± 2 dBrn, 10 dBrn to 20 dBrn
	± 1 dBrn, 20 dBrn to 90 dBrn

Detector: True RMS
Filters: C-message, 3 kHz flat, 15 kHz flat

2.6.3.2 Noise With Tone Measurements.
Transmitter Mode: holding tone.
Receiver:

(a) Level Range	10 dBrn to 90 dBrn
(b) Resolution	0.1 dB
(c) Resolution	1 dB
(d) Accuracy	± 2 dBrn, 10 dBrn to 20 dBrn
	± 1 dBrn, 20 dBrn to 90 dBrn

Detector: True RMS
Filters: C-notched, 3 kHz Flat-notched

2.6.3.3 Signal-to-Total Distortion (S/TD) Ratio.
Transmitter Mode: Holding tone
Measurement:

(a) Range	10 dB to 45 dB
(b) Resolution	1 dB
(c) Holding Tone Level	-40 dBm to +10 dBm
(d) Accuracy	± 1 dB (10 dB to 30 dB)
	± 2 dB (31 dB to 45 dB)

Detector: True RMS
Filters: C-notched

2.6.3.4 Transients.
Transmitter Mode: holding tone
Receiver:

(a) Holding Tone Level	-40 dBm to +10 dBm
(b) Filter	C-message, C-notched
(c) Count Range	0 to 998 or greater for each level
(d) Count rate	8 counts per second ± 10%
(e) Timer	
Interval	0 to 60 minutes or more in less 1 min or less steps
Accuracy	± 5% or better

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2.6.3.4.1 Three-level Impulse Noise.

Threshold reference setting:

(a) Range	30 dBrn to 104 dBrn in 1 dB steps
(b) Accuracy	± 1 dB
(c) Threshold Delta	3 levels min., at least one delta setting of 4 dB
(d) Blanking Interval	125 ms ± 10 %

2.6.3.4.2 Phase Hits Measurement.

Threshold reference setting:

(a) Range	5 degrees to 45 degrees in 5 degree steps
(b) Accuracy	± 0.5 degrees ± 10% of setting
(c) Transition time	greater than 4 ms ± 0.5 ms
(d) Blanking Interval	125 ms ± 10 %

2.6.3.4.3 Gain Hits Measurement.

Threshold reference setting:

(a) Range	2 dB, 3dB, 6dB
(b) Accuracy	± 0.5 dB
(c) Transition time	greater than 4 ms ± 0.5 ms
(d) Blanking Interval	125 ms ± 10 %

2.6.3.4.4 Dropout Measurement.

Threshold reference setting:

(a) Range	12 dB below holding tone
(b) Accuracy	± 1 dB
(c) Transition time	greater than 4 ms ± 0.5 ms

2.6.3.5 Phase Jitter and Amplitude Jitter Measurements.

Transmitter Mode: holding tone

(a) Holding Tone Level	-40 dBm to +10 dBm
(b) Holding Tone Freq.	990 to 1030 Hz
(c) Weighting Filters	20-300 Hz, 4-300 Hz

Amplitude Jitter:

(a) Range	0.0 to 25.0 %
(b) Resolution	0.1 %
(c) Accuracy	± 0.2% ± 5% of reading

Phase Jitter:

(a) Range	0.0 to 25.0 degrees
(b) Resolution	0.1 degrees
(c) Accuracy	± 0.2 degrees ± 5% of reading

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2.6.3.6 Peak-to-Average Ratio (P/AR) Measurements. P/AR is defined in IEEE 743-1984. IEEE 743-1995 refers to IEEE 743-1984. P/AR is a measure of intersymbol interference in voiceband data signals. P/AR is designed to evaluate the simultaneous effect of envelope delay distortion, bandwidth reduction, and poor return loss (gain and phase ripples) on high-speed data transmission. IEEE 743-1995 equivalent measurements are acceptable.

Transmitter:	
(a) Range	-40 dBm to 0 dBm
(b) Frequency	16 tones between 140 Hz and 3891 Hz [Fn = (2n-1) +15.625, for n=1 to 16]
(c) Resolution	1.0 dB
Receiver:	
(a) Level Range	-40 dBm to 0 dBm, true RMS
(b) P/AR Range	0 to 120 units
(c) Resolution	1 P/AR unit
(d) Accuracy	± 2 P/AR units (from 40 to 110 P/AR units) ± 4 P/AR units (from 0 to 120 P/AR units)
Detectors:	full-wave rectified average and absolute peak

2.6.3.7 Impedances.

(a) Terminated:	600 Ω and 900 Ω
(b) Bridged:	> 25,000 Ω

2.6.3.8 Bridging Loss. Bridging loss shall be less than 0.2 dB.

2.6.4 Connectors. The TIMS equipment shall provide the following panel connectors: WECO type 310 or 309 dual bantam jacks and a modular phone jack (RJ11C type).

2.6.5 Dial and Talk. The TIMS shall provide built-in dialing capabilities for pulse and tone dialing. The TIMS shall provide a built-in microphone, a built-in speaker, and an earphone jack

2.6.6 Accessories. The TIMS shall be delivered with a ruggedized transit case, batteries, and AC adapter. The transit case shall have provisions for the stowage of accessories. Stowage shall be contained in a manner that prevents damage to the accessories or equipment during transit.

2.6.7 Printer Interface. The instrument shall be provided with an EIA/TIA-232-F interface or other instrument interface compatible with a printer.

2.6.8 Physical Characteristics. The equipment shall be portable with the following physical characteristics.

2.6.8.1 Dimensions. The height, width, and depth of the equipment shall be compatible with hand-held operation. The equipment shall be within 5 inches wide, 8 inches high, and 2 inches deep.

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2.6.8.2 Weight. The weight of the equipment shall be compatible with hand-held operation. The total weight of the equipment, excluding accessories and manuals, shall not exceed 11 pounds (5 kilograms).

2.6.9 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800 paragraph 6.6, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS

3.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 Institute of Electrical and Electronics Engineers (IEEE) standards products may be obtained at http://www.ieee.org or available from IEEE Operations Center. Phone: (800)678-IEEE. Mail: IEEE Operations Center, Sales Office, 445 Hoes Lane, PO Box 1331, Piscataway, NJ, 08855-1331, USA.

6.1.3 FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dpap/dars/dfars/index.htm .

6.2 Key Words.

Hand-held
Peak-to-Average-Ratio
Signal-to-Noise Ratio

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
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